

Generative AI Core Concepts & Architecture / 生成式 AI 核心概念與架構

Comprehensive Reference Guide • English & Traditional Chinese (繁體中文) Bilingual Version

1. Model Parameters / 模型參數設定

1. Max Response / Max Tokens (最大生成長度)

ENG: Controls the maximum length of the generated text by setting a hard limit on tokens. If configured too low, responses may be cut off mid-sentence.

CHI: 透過設定 Token 數量的硬性上限，來控制生成文本的最大長度。如果設定過低，AI 的回覆可能會在中途或句子未完時中斷。

2. Temperature (溫度)

ENG: Controls the randomness and creativity of the output (not verbosity). Low temperature (e.g., 0.2) yields predictable, factual responses; high temperature (e.g., 0.9) produces creative, diverse outputs.

CHI: 控制輸出的隨機性與創意度（而非字數多寡）。較低的溫度（例如 0.2）會產生高度可預測且著重事實的回覆；較高的溫度（例如 0.9）則會帶來更具創意與多樣化的輸出。

3. Stop Sequence (停止序列)

ENG: Instructs the model to immediately cease text generation once a specific character string or token (such as `\n` or `[END]`) appears. Essential for preventing the model from simulating user turns or rambling.

CHI: 指示模型在出現特定的字元序列或符號（例如 `[END]` 或換行符號 `\n`）時，立即停止生成文本。這對於保持輸出格式整潔及防止聊天機器人自我扮演非常實用。

4. Presence Penalty (存在懲罰)

ENG: Applies a penalty based on whether a token has already appeared in the generated text, encouraging the model to introduce entirely new subjects and diverse concepts rather than repeating themes. Range: -2.0 to 2.0.

CHI: 根據某個詞彙是否已在已生成的文本中出現過來實施懲罰，以此激勵模型引入全新的主題與概念，避免在同一個話題上原地兜圈子。數值範圍通常為 -2.0 至 2.0。

5. Past Messages Included (包含歷史訊息 / 上下文範圍)

ENG: Manages short-term memory inside the conversation context window. It determines how many previous back-and-forth interactions are bundled and re-sent to the AI to maintain continuous contextual awareness.

CHI: 管理上下文視窗內的短期記憶。它決定了有多少條歷史對話會被打包並隨新問題一起重新發送給 AI，從而使模型能夠理解並維持上文下理的連貫性。

2. Building AI Agents / 建立 AI 代理系統

ENG: Software applications built on generative AI that use natural language to reason, utilize tools to automate tasks, and perform multi-step, goal-oriented workflows.

CHI: 以生成式 AI 為基礎的軟體應用程式。它們能使用自然語言進行推理、利用各種工具實現自動化運作，並獨立執行多步驟、目標導向的工作流程。

Three Pillars of an Agent / 代理系統三大核心支柱

- **Model (Brain / 大腦):**

ENG: The generative AI model used for linguistic understanding and strategic reasoning.

CHI: 用於語言理解、情境分析和邏輯推理的生成式 AI 模型。

- **Instructions (Role / 角色):**

ENG: The system prompt that defines the agent's behavior, persona, style, and operational constraints.

CHI: 定義代理人身份、行為守則、回應風格以及邏輯限制的系統提示詞 (System Prompt)。

- **Tools (Action / 行動):**

ENG: Callable capabilities that allow the agent to interact, execute tasks, and modify external environments.

CHI: 可供調用的外部功能，賦予代理人與外界互動、執行具體任務並改變外部資料環境的能力。

Microsoft Foundry Tools Management / 工具庫集中管理

Foundry centrally coordinates tools via a unified **Tool Catalog (工具目錄)**:

- **Knowledge Tools (知識工具):**

ENG: Provide secure access to external information silos (e.g., web search engines, corporate document databases).

CHI: 提供外部資訊的訪問管道 (例如網頁搜尋引擎、企業文檔數據庫)。

- **Action Tools (行動工具):**

ENG: Enable specialized tasks to be performed directly (e.g., sending automated emails, updating calendars, calling custom REST APIs).

CHI: 允許代理人直接執行特定任務 (例如發送電子郵件、更新行事曆、調用自訂 API)。

Retrieval-Augmented Generation (RAG) Flow / 檢索增強生成流程

ENG: Technique to add context to the prompt, allowing models to answer questions about specific data not included in their original training.

CHI: 一種為提示詞增加背景資料的技術，使模型能夠針對其原始訓練資料中未包含的特定私有數據進行精準回答。



Grounding & Accuracy (知識落地與準確度):

- *Domain Specificity* (領域專一性): Prevents generic hallucinations by forcing reliance on authorized data. / 強迫模型依賴授權數據，防止胡言亂語。
- *Traceability* (可追溯性): Responses include citations (e.g., "Source: HR_Policy.pdf"). / 回覆包含引用來源，提升透明度。
- *Microsoft Foundry IQ*: Handles chunking and vector embeddings automatically. / 自動化處理文件切片與向量化嵌入，並可在多代理之間高效共享。

Multi-Agent Systems & MCP / 多代理系統與模型上下文協議

Instead of one monolith doing everything, specialized agents collaborate across structural workflows. Coordination is standardized via the **Model Context Protocol (MCP)**, an open universal adapter architecture separating concerns between the **MCP Client** (Agent requesting insights) and the **MCP Server** (Service running the processing logic).

與其依靠單一主體包辦所有工作，多個**專業化代理人**可在 workflow 中協同配合。這種協調由 **Model Context Protocol (MCP，模型上下文協議)** 實現標準化，該協議將 **MCP 客戶端**（發出請求的代理）與 **MCP 伺服器端**（執行功能的託管服務）的關注點進行清晰分離。

3. Performance and Scale / 效能與大規模擴展

Infrastructure resource management policies under heavy production traffic conditions. / 當面對高流量生產環境時的底層資源管理機制。

Tokens Per Minute (TPM) Allocation / 每分鐘 Token 分配量:

ENG: The velocity and scale rating at which a hosted deployment can process word inputs and outputs. It dictates strict performance ceilings.

CHI: 衡量一個部署模型每分鐘能處理輸入與輸出文本的速度與規模指標。它決定了系統的效能硬性上限。

Capacity / 承載容量:

ENG: Scales upwards when TPM limits are higher, enabling the infrastructure cluster to safely absorb simultaneous traffic spikes across concurrent sessions.

CHI: 當分配的 TPM 越高時，系統的承載力便隨之增加，使雲端基礎設施能夠平穩處理大量併發、同時發生的用戶請求。

Max Output Token Reservation / 最大輸出保留:

ENG: The backend reserves the full potential output token volume in TPM space the exact millisecond a request arrives, preserving structural query stability mid-generation.

CHI: 為了確保每次互動的穩定性，系統會在收到請求的瞬間，在 TPM 額度中預留該請求對應的最大輸出空間，防止回覆中途因他人搶佔資源而斷線。

Throttling / 節流與限速保護:

ENG: A defensive protective framework that intentionally retards heavy computing tasks or delays arrivals when a cluster reaches its physical scaling threshold.

CHI: 一種自我防禦性的保護機制。當系統達到運算極限時，會刻意放慢處理速度或延後新請求（例如傳回 HTTP 429），以維護整個伺服器叢集的完全穩定，避免崩潰。